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Three-dimensional analysis of shape rolling using a generalized upper bound approach

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ABSTRACT

A generalized analytical method has been proposed to study external shape and calculate pressure and torque for the process of rolling shaped sections. The proposed method is based on the upper bound analysis. To determine kinematically admissible velocity fields, flow lines have been mathematically formulated using the parameterization of curves. Based on the calculated velocity fields, an upper bound has been found on rolling power. The calculated power has been minimized with respect to unknown variables to yield the value closest to the actual power required.

The proposed general method has been applied to certain rolling passes including square-to-oval and round-to-oval. Velocity fields and power relations have been obtained for each process. Computer analysis has been carried out for all the mentioned passes using dimensions and conditions from experiments conducted by other workers. External shape, average roll torque and average rolling pressure results from the analysis have been compared with analytical solutions, numerical analyses and experimental data presented by other workers. It was concluded that the present method gave very good results and was quicker and easier to use.

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1. Introduction

The primary purpose in analyzing metal forming operations is to determine unknown process variables in order to decide the best design of die/work piece to achieve optimum results. In the case of shape rolling, the purpose becomes optimum design of rolls and the roll pass sequence.

However, as shape rolling operations involve large numbers of process variables and include a complex three-dimensional material flow, the design of pass sequences and the prediction of roll load and torque, entry and exit velocities have mostly been based on empirical formulae developed for the most conventional materials and rolling conditions (Wusatowski, 1969; Lange, 1985).

Still, as new shapes and materials are introduced to the rolling industry, development of empirical solutions for roll pass sequences as well as other process parameters requires extensive experiments. As a result, development of theoretical-based methods to analyze external deformation, exit velocity, roll torque, pressure and force for the rolling of shaped sections has become extremely attractive.

One of the first theoretical works was a numerical approach to the three-dimensional plastic deformation of rolling to investigate the side spread in flat rolling by Oh and Kobayashi (1975). They conducted the pioneering study in this field by applying an extremum principle for rigid, perfectly plastic materials combined with numerical analysis. Since then, several other researchers have used three-dimensional finite

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Nomenclature

a, b	half-length of major and minor axes, respectively, in the exit cross-section of square-to-oval and round-to-oval rolling passes
a_1, b_1, c_1, a_2, b_2, c_2	arbitrary coefficients for the auxiliary functions of Bezier control points
A_1	entry cross-section
A_2	exit cross-section
A_f	roll–workpiece interface area
f, g, h	functions of u, q, t representing position of control points in Cartesian coordinates
$F_0(u, q)$, $G_0(u, q)$, $H_0(u, q)$, $F_1(u, q)$, $G_1(u, q)$, $H_1(u, q)$	auxiliary functions introduced to formulate control points in Bezier curves
J	Jacobian for conversion of coordinates from x, y, z to u, q, t
m	friction shear factor
$\vec{n}$	unit vector normal to the roll surface at roll–workpiece interface
$\vec{N}$	vector normal to the roll surface at roll–workpiece interface
Q	reduction of cross-sectional area during rolling
r	diameter of billet cross-section in the round-to-oval pass
$\vec{r}(t)$	position vector of any given point on Bezier curve
s	half-length of billet cross-section in a square-to-oval rolling pass
u, q, t	parameters introduced to enable definition of deformation zone streamlines as curves
V_x, V_y, V_z	components of the three-dimensional velocity vector
V_e	billet velocity upon entry to the deformation zone
V_f	velocity discontinuity at roll–workpiece interface
V_r	tangential roll velocity
V_1	velocity discontinuity at entry
V_2	velocity discontinuity at exit
$\vec{V}_i$	position vectors of control points on Bezier curves
W	total power required to deform the rolling material
W_e	power required to overcome shear forces resulting from velocity discontinuity at entry
W_f	power required to overcome friction
W_i	power required for plastic deformation
W_x	power required to overcome shear forces resulting from velocity discontinuity at exit
y_c	distance from the designated coordinate reference to upper roll axis
z_0	length of the deformation zone

Greek symbols

α	angle between vector normal to the roll surface at roll–workpiece interface and the y-axis
$\dot{\varepsilon}_{ij}$	strain rates in tensor notation
$\dot{\bar{\varepsilon}}$	effective strain rate
θ	angle between X or Y axes and the arbitrary end point of a streamline at exit cross-section
φ	angle between X or Y axes and the arbitrary start point of a streamline at entry cross-section
$\bar{\sigma}$	average flow stress

element method – FEM – for analyzing spread in rolling. Kim et al. (1992) found a computationally efficient slab and finite element method for the single-pass rolling of a square billet to an oval shape and Lee et al. (1997), used the same method to investigate two multi-pass square-to-round sequences.

Park and Oh (1990) developed a rigid-plastic 3D FEM code for the analysis of shape rolling processes, which is capable of handling arbitrary cross-sections. Mori and Osakada (1991, 1993) developed a rigid-plastic 3D FEM to simulate steady-state deformation in shape rolling with grooved rolls. Kim et al. (1991) combined the two-dimensional rigid-plastic finite element method with the slab method to reduce the amount of computation without losing accuracy.

Saito et al. (1984) proposed an upper bound method that was capable of predicting lateral spread, elongation and thickness in flat, concave (diamond) and convex (box) passes.

Lee and Kim (2001) presented a semi-analytical method for the approximate estimation of roll force in a round–oval–round pass sequence. They introduced the concept of weak plane-strain deformation taking into account the deformation characteristics of the workpiece when the equation for roll force was formulated. Bayoumi (1998) used a flow-line field solution to analyze material flow and stresses in the steady-state rolling of a round–oval–round sequence. Hsiang and Lin (2001) developed a piecewise complex model combining the slab and finite element methods. They divided the deformation zone into several slabs and used FEM to calculate sequentially the velocity field, metal flow, stress and strain distribution for each slab. Kennedy (1986) developed an upper bound method for the analysis of metal deformation and stresses in shape rolling of geometrical profiles without protrusions based on the general velocity fields proposed by Hill (1963).

Kim and Im (2002) developed a three-dimensional finite element program for analyzing different shape rolling processes based on the rigid thermo-viscoplastic approach which also considered the effects of heat transfer. The developed program was used to simulate square-to-oval, round-to-oval, and square-to-diamond passes as well as the rolling of an H-beam at different friction conditions with and without temperature effects. Deformed shape, stress and temperature distributions in the workpiece, and forming loads were investigated.

Behzadipour et al. (2004) introduced a neural network-based model to estimate power in flat, and shape, rolling. The model used experimental and analytical data from flat rolling for training the network and introduced a shape change quantification factor to enable application to shape rolling.

Milenin et al. (2004) proposed a mathematical model to address the problem of non-isothermal metal deformation during multi-pass rolling. The model was applied to the rolling of a steel angle bar in six passes and a double-pass round–oval–round rolling sequence.

Capece Minutolo et al. (2005) used the finite element technique to analyze spread, surface profile and cross-sectional area of the workpiece in round-to-oval and round–flat–oval passes in the rolling of steel. Their work included comparisons with an analytical method for the prediction of the mean effective strain of the workpiece during shape rolling as well as with experimental data such as maximum spread and radius.

This study attempts to present a method for the analysis of rolling operations which involve simple geometrical shapes such as squares, ovals, rounds, etc. The analysis uses the upper bound solution, assumes elastic deformations to be negligible, and does not consider combined thermo-mechanical effects. Strain rates and velocities are derived from the parametric definition of streamlines in the deformation zone. Solutions are applied to the square-to-oval and round-to-oval rolling operations.

2. Method of analysis

The method proposed in the present study is based on the upper bound method of analysis (Johnson and Mellor, 1983). The first step in applying the upper bound method is the identification of kinematically admissible velocity fields through an assumption of the shape of the streamlines in the deformation zone. Based on this velocity field, the total power dissipation for the metal forming process is calculated. Since this predicted power is inherently higher than the actual value required for the process, the shape of streamlines and metal flow distribution can be determined by minimizing the power dissipation with respect to unknown variables of the velocity field.

Fig. 1 shows a quarter of the deformation zone considered for the shape rolling of an arbitrary section to another arbitrary section.

The geometrical configuration shown in Fig. 1 is based on the choice of a streamline BB' and a stream surface $OAA'O'$. The entry and exit sections are bound by planes represented by OCD and $O'C'D'$, respectively. To simplify relations, a left-hand coordinate system has been employed. It has been assumed that the deformation zone consists of such stream surfaces as $OAA'O'$ within which streamlines such as BB' are located. It has been further assumed that any given particle P in the deformation zone travels on such streamlines and the proportionality relation $OB/OA = O'B'/O'A'$ holds along the entire trajectory. It can thus be seen that this assumption presents a method to mathematically define the deformation zone in the shape rolling of an arbitrary section to another arbitrary section (Abrinia, 1990).

Bezier's parametric curve formulation has been used to describe the streamlines (Qiulin and Davies, 1987). Bezier used Bernstein polynomials as blending functions for the vectors. He also suggested the use of control points, where a curve of degree n has $n+1$ control points V_i ($i = 0, 1, 2, 3, \ldots, n$). The Bezier curve passes only through the first and final control points and approximates towards the intermediate points. This research uses cubic Bezier curves. As illustrated in Fig. 2, the curve so formulated passes through points V_0 and V_3, and has tangents at V_0 from V_0 to V_1 and at V_3 from V_2 to V_3. It is also evident from the figure that as one or more control points are moved, the shape of the Bezier curve changes accordingly.

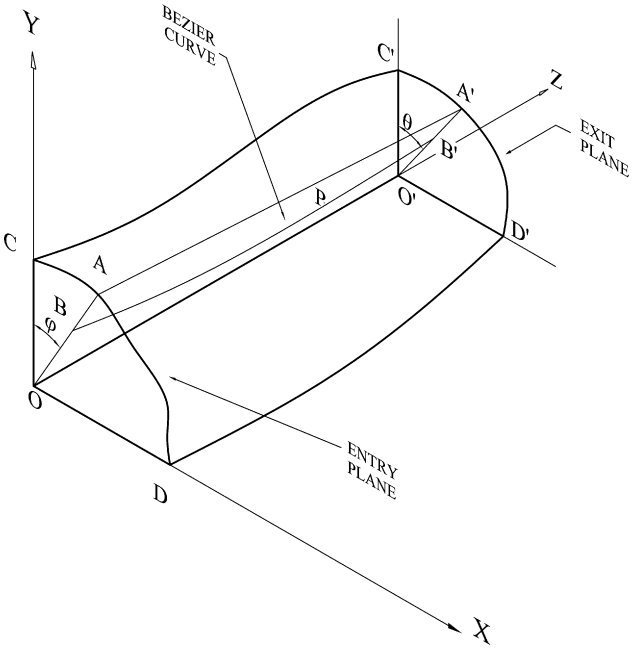


Fig. 1 – A quarter of the deformation zone assumed for the rolling of an arbitrary shape to another.

Bezier's curve of degree 3 is defined as:

$$\vec{r}(t) = (1-t)^3\vec{V}_0 + 3t(1-t)^2\vec{V}_1 + 3t^2(1-t)\vec{V}_2 + t^3\vec{V}_3 \quad (1)$$

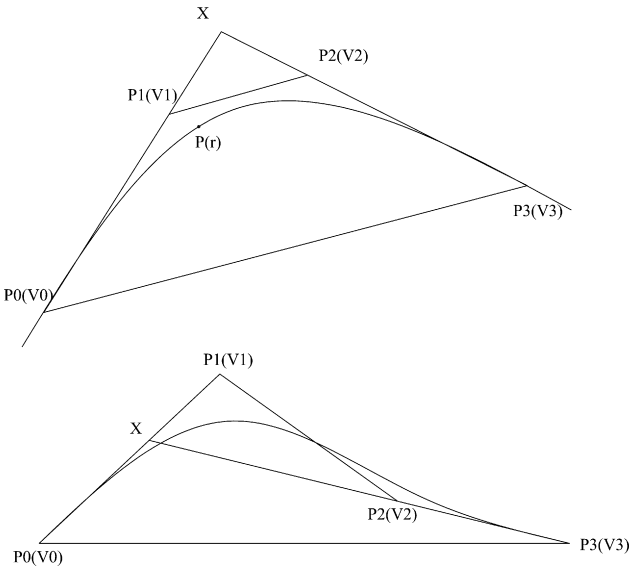


Fig. 2 – A typical Bezier curve of degree 3 passes through V_0 and V_3 and has tangents at V_0 from V_0 to V_1 and at V_3 from V_2 to V_3.

In matrix formation:

$$\vec{r}(t) = \begin{bmatrix} 1 & t & t^2 & t^3 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ -3 & 3 & 0 & 0 \\ 3 & -6 & 3 & 0 \\ -1 & 3 & -3 & 1 \end{bmatrix} \begin{bmatrix} \vec{V}_0 \\ \vec{V}_1 \\ \vec{V}_2 \\ \vec{V}_3 \end{bmatrix} \quad (2)$$

where $\vec{V}_0$ through $\vec{V}_3$ mark the position vectors of the control points, $0 \le t \le 1$ is the main curve defining parameter and $\vec{r}(t)$ is the position vector of a given point on the curve.

In order to express streamlines within the deformation zone as Bezier curves, position vectors $\vec{V}_0, \vec{V}_1, \vec{V}_2$ and $\vec{V}_3$ must be appropriately defined. For the sake of generality and in order to find relations applicable to the majority of shaped sections in industrial rolling operations, the following general format has been chosen for the vectors:

$$\begin{aligned} \vec{V}_0 &= F_0(u,q)\vec{i} + G_0(u,q)\vec{j} \\ \vec{V}_1 &= a_1F_0(u,q)\vec{i} + b_1G_0(u,q)\vec{j} + c_1H_0(u,q)\vec{k} \\ \vec{V}_2 &= a_2F_1(u,q)\vec{i} + b_2G_1(u,q)\vec{j} + c_2H_1(u,q)\vec{k} \\ \vec{V}_3 &= F_1(u,q)\vec{i} + G_1(u,q)\vec{j} + H_1(u,q)\vec{k} \end{aligned} \quad (3)$$

u and q are two parameters that vary between 0 and 1 and control the position of vector components in the entry and exit planes. In the subsequent section, these parameters have been given specific definition for each rolling shape under study. F_0 and G_0 are functions of u and q that determine the location of point B (Fig. 1) and its position vector $\vec{V}_0$. In each rolling section subsequently studied, these functions have been defined to enable $\vec{V}_0$ to encompass the entire entry cross-section. F_1, G_1 and H_1 determine the location of B' and position vector $\vec{V}_3$. The two intermediate control points have been defined using these same functions with H_0 added for $\vec{V}_1$ and arbitrary coefficients a_1, a_2, b_1, b_2, c_1 and c_2 introduced to adjust the position of intermediate control points. As depicted in Fig. 2, $\vec{V}_0$ and $\vec{V}_3$ are the position vectors of points B and B', respectively. Combining Eqs. (1) and (3):

$$\vec{r} = f(u,q,t)\vec{i} + g(u,q,t)\vec{j} + h(u,q,t)\vec{k} \quad (4)$$

and in the Cartesian coordinate system:

$$\begin{aligned} x &= f(u,q,t) \\ y &= g(u,q,t) \\ z &= h(u,q,t) \end{aligned} \quad (5)$$

The total power W required to deform the material in a rolling operation is expressed as the sum of the following components: the power required to deform the material plastically, W_i; the power required to overcome shear forces resulting from velocity discontinuities at entry W_e, at exit W_x; and the power required to overcome friction, W_f. In mathematical terms:

$$W = W_i + W_e + W_x + W_f \quad (6)$$

where:

$$\begin{aligned} W_i &= \bar{\sigma}\int_V \left(\frac{2}{3}\dot{\varepsilon}_{ij}.\dot{\varepsilon}_{ij}\right)^{1/2}.dV \\ W_e &= \frac{\bar{\sigma}}{\sqrt{3}}\int_{A_1} V_1.dA_1 \\ Wx &= \frac{\bar{\sigma}}{\sqrt{3}}\int_{A_2} V_2.dA_2 \\ W_f &= \frac{m.\bar{\sigma}}{\sqrt{3}}\int_{A_f} V_f.dA_f \end{aligned} \quad (7)$$

Assuming the deforming material flows according to the Levy–Mises rule and obeys the Mises yield criteria (Johnson and Mellor, 1983), incompressibility condition in three-dimensional metal flow can be expressed as:

$$\frac{\partial V_x}{\partial x} + \frac{\partial V_y}{\partial y} + \frac{\partial V_z}{\partial z} = 0 \quad (8)$$

V_x, V_y and V_z are the components of the three-dimensional velocity vector along coordinate axes X, Y and Z, respectively. Equating the unit tangent vector for the point P on the streamline BB' (Fig. 1) to the unit velocity vector yields:

$$\begin{aligned} V_x &= \frac{f_t}{h_t}V_z \\ V_y &= \frac{g_t}{h_t}V_z \\ V_z &= M(u,q,t) \end{aligned} \quad (9)$$

A kinematically admissible velocity field must essentially satisfy the incompressibility condition. Replacing V_x, V_y and V_z from (9) into (8) and solving for $M(u, q, t)$:

$$M = \frac{C \cdot h_t}{h_t(f_u g_q - f_q g_u) + h_q(f_t g_u - f_u g_t) + h_u(f_q g_t - f_t g_q)} \quad (10)$$

where C is a constant to be determined from the specific boundary conditions of each section studied (Appendix A contains the detailed procedure for the derivation of Eq. (10)).

Velocities calculated using (10) automatically satisfy the incompressibility or the volume constancy condition.

Using the parametric notation employed in this study, the plastic deformation component of the total rolling power may be expressed as follows:

$$W_i = \frac{2\bar{\sigma}}{\sqrt{3}}\int_V \left(\frac{\dot{\varepsilon}_{xx}^2 + \dot{\varepsilon}_{yy}^2 + \dot{\varepsilon}_{zz}^2}{2} + \dot{\varepsilon}_{xy}^2 + \dot{\varepsilon}_{yz}^2 + \dot{\varepsilon}_{zx}^2\right)^{1/2} |J| \cdot du\, dq\, dt \quad (11)$$

$\dot{\varepsilon}_{xx}$ ... are strain rates in various directions, $\bar{\sigma}$ is the flow stress and $|J|$ is the determinant of the Jacobian for transformation of coordinates from x, y, z to u, q, t.

The velocity discontinuity components of total rolling power can be expressed as follows. At entry:

$$W_e = \frac{\bar{\sigma}}{\sqrt{3}}\int_0^1\int_0^1 (V_x^2 + V_y^2)_{t=0}^{1/2} \frac{\partial(x,y)}{\partial(u,q)}_{t=0} du\, dq \quad (12)$$

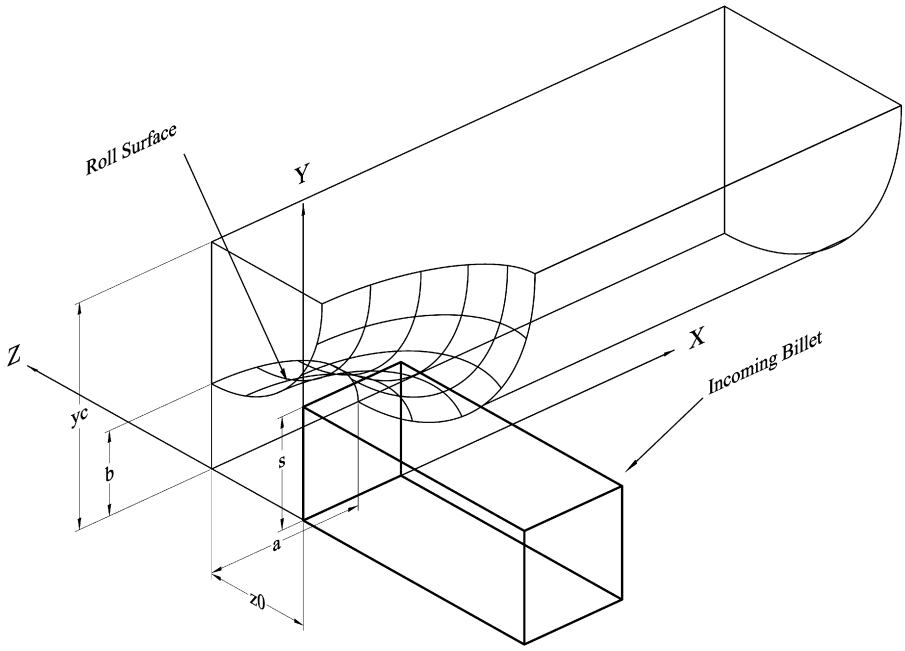


Fig. 3 – A quarter of the deformation zone for square-to-oval pass (depicted in sections for clarity).

and at exit:

$$W_x = \frac{\bar{\sigma}}{\sqrt{3}} \int_0^1 \int_0^1 (V_x^2 + V_y^2)_{t=1}^{1/2} \frac{\partial(x, y)}{\partial(u, q)}_{t=1} du\, dq \quad (13)$$

The friction component:

$$W_f = \frac{m\bar{\sigma}}{\sqrt{3}} \int_0^1 \int_0^1 \sqrt{(V_t - V_r)^2 + V_y^2}_{u=1} \sec\alpha \frac{\partial(x, z)}{\partial(q, t)}_{u=1} dq\, dt \quad (14)$$

where $\sec\alpha = \sqrt{(N_1^2 + N_2^2 + N_3^2)}/N_2$ and $\vec{N} = (\partial\vec{r}/\partial q_{u=1}) (\partial\vec{r}/\partial t_{u=1})$, $V_t = \sqrt{V_x^2 + V_z^2}$ and V_r is the tangential roll velocity.

3. Simulation of pass sequences

The analytical model presented earlier has been employed to simulate specific passes in shape rolling as follows.

3.1. *Square-to-oval pass*

Fig. 3 shows a quarter of the three-dimensional deformation zone for the rolling of a square billet to an oval bar. Lower roll has been omitted for clarity. As the rolling geometry is assumed to be symmetrical, only a quarter of the deformation zone suffices for the purpose of this analysis.

It has been assumed that solid grooved rolls draw the billet inside the roll gap and reduce its cross-section. The projected length of the deformation zone z_0 (Fig. 3) is calculated using the geometrical specifics of the pass and billet dimensions.

The analysis requires an analytical representation of the three-dimensional roll groove surface. Assuming the curve defining the exit cross-section of the rolls has been defined as $y = f(x)$, every point on the roll groove surface is generated by the rotation of a corresponding point on the curve about the upper roll axis $y = y_c$. The three-dimensional roll groove surface may thus be expressed as

$$(y_c - f(x))^2 = (y - y_c)^2 + (z_0 - z)^2 \quad (15)$$

y and z are the coordinates of a rotated point on the roll groove surface.

The entry cross-section for the square-to-oval pass has been assumed to be a square with sides 2*s* in length and the exit cross-section an oval with a major axis of length 2*a* and a minor axis of length 2*b*.

Considering the complexity of the deformation zone and the approximate metal flow in the square-to-oval pass, it was decided for the purpose of this study that a single Bezier curve will be insufficient to describe the entire deformation zone. To facilitate the analysis, the following assumptions have been made.

- That the deformation zone consists of three smaller zones; the zone bound between the plane *OBB′O′*and the surface *OAB′O′*, the zone limited to *OAB′O′*and the surface *OAA′O′*, and the zone lying between *OAA′O′* and the surface *CAC′*, as shown in Fig. 4.
- That a particle entering the deformation zone from within the perimeter of triangle *OAB* travels along its assigned streamline, e.g. *QQ′*, which lies somewhere between *OBB′O′*and *OAB′O′* until it reaches *OAB′O′*, where a second streamline, e.g. *Q′Q″*, originates. The particle then resumes its course along the latter which lies between *OAB′O′* and the plane *OAA′O′* (see Fig. 4).
- That particles entering the deformation zone from within the perimeter of triangle *OAC* travel along designated streamlines, e.g. *SS′*, without encountering a second surface

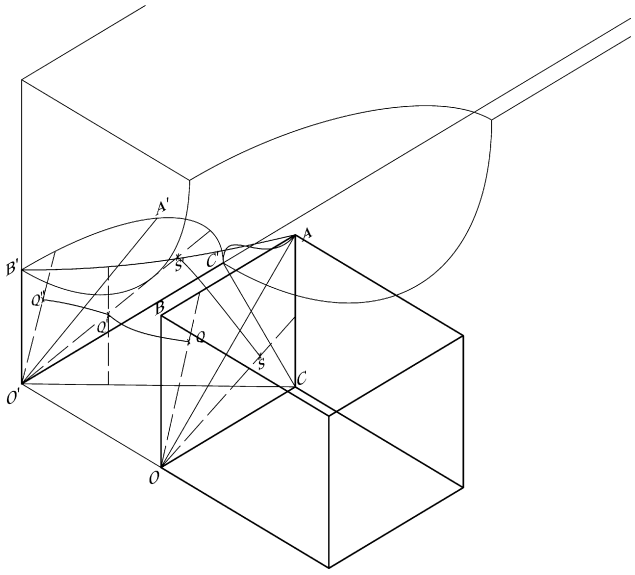


Fig. 4 – Zoning of the deformation region in square-to-oval pass. Typical streamlines have been depicted as QQ″ and SS′.

to exit the deformation zone from the area bound by $O'A'$ and $O'C'$ (see Fig. 4).

- That the deformation zone can be aptly described by the three typical streamlines mentioned; QQ', $Q'Q''$ describe the first half and SS' describes the second half (see Fig. 4).

Therefore, for each of the three sections of the deformation zone in the square-to-oval rolling pass, a separate Bezier curve has been formulated and extra parameters have been introduced to ensure continuity of slope and curvature in transition from one curve to another. More details concerning the parametric Bezier formulation for each curve and the resulting velocity field for each zone are given in Appendix B.

The sum of the three curves provides a parametric description of all the streamlines in the deformation zone. The summation includes 15 unknown coefficients a_1 to c_2, k_1, k_2, a'_1 to c'_2 and V_e – see Appendix B – which will be determined through optimization of the total power required with the velocity field calculated.

A MATLAB computer code utilizing the Genetic Algorithm was generated to perform the optimization. Using Simpson's Three Point Rule, W_i was calculated by numerical integration with respect to each of the parameters u, q and t. For W_e and W_x analytical integration with respect to u was followed by numerical integration with respect to q using a three-point Simpson Rule. For W_f, integration was performed numerically with respect to both parameters using the five-point rule.

3.2. *Round-to-oval pass*

Fig. 5 shows a quarter of the typical deformation zone in the round-to-oval pass.

For the round-to-oval pass, it has been assumed that the entry cross-section is a circle of radius r and the exit section an oval with a major axis of length $2a$ and a minor axis of length $2b$ (Fig. 5).

Due to the relatively less complex metal flow patterns in round-to-oval rolling as compared to the square-to-oval rolling sequence, a single Bezier curve was found sufficient to describe the entire deformation zone. The Bezier control

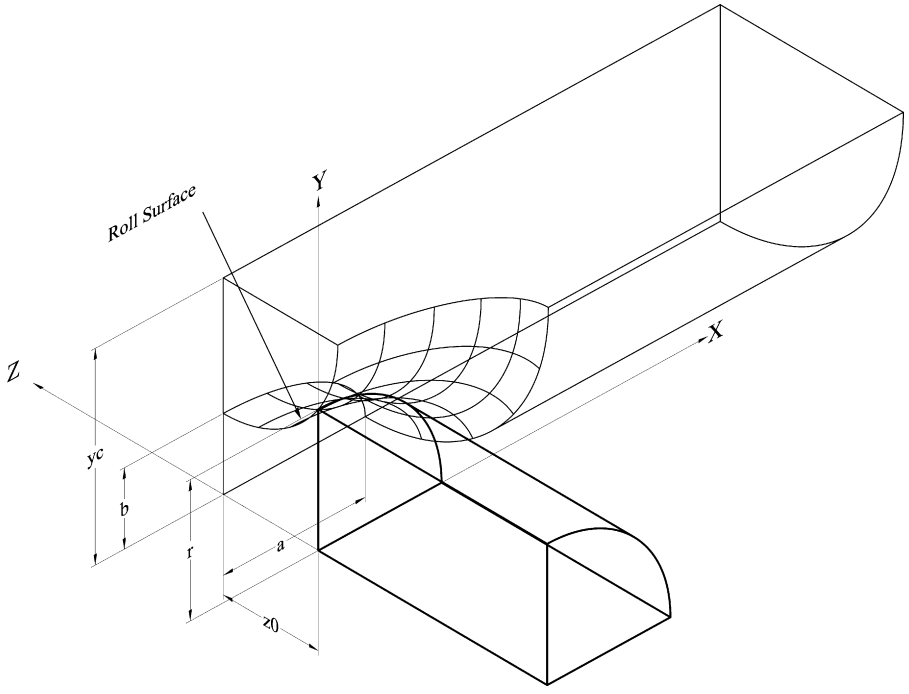


Fig. 5 – A quarter of the deformation zone for round-to-oval pass (depicted in sections for clarity).

points have been found as follows:

$$\begin{aligned}\vec{V}_0 &= ur\,\sin\varphi\cdot\vec{i} + ur\,\cos\varphi\cdot\vec{j}\\ \vec{V}_1 &= a_1 ur\,\sin\varphi\cdot\vec{i} + b_1 ur\,\cos\varphi\cdot\vec{j} + c_1 z_0\cdot\vec{k}\\ \vec{V}_2 &= a_2 ua\,\sin\varphi\cdot\vec{i} + b_2 ub\,\cos\varphi\cdot\vec{j} + c_2 z_0\cdot\vec{k}\\ \vec{V}_3 &= ua\,\sin\varphi\cdot\vec{i} + ub\,\cos\varphi\cdot\vec{j} + z_0\cdot\vec{k}\end{aligned} \tag{16}$$

where $\varphi = \pi q/2$ and the curve itself in parametric terms:

$$\vec{r}(u,q,t) = (1-t)^3\vec{V}_0 + 3t(1-t^2)\vec{V}_1 + 3t^2(1-t)\vec{V}_2 + t^3\vec{V}_3$$

and in Cartesian coordinates:

$$\vec{r}(u,q,t) = f(u,q,t)\vec{i} + g(u,q,t)\vec{j} + h(u,q,t)\vec{k}$$

Velocities in the three coordinate directions may be expressed as:

$$\begin{aligned}M &= V_z\\ &= \frac{\pi V_e r^2 u}{2[(f_q g_u - f_u g_q) - (h_q/h_t)(f_t g_u - f_u g_t) - (h_u/h_t)(f_q g_t - f_t g_q)]}\\ V_x &= \frac{f_t}{h_t}V_z\\ V_y &= \frac{g_t}{h_t}V_z\end{aligned} \tag{17}$$

The velocities derived satisfy both the incompressibility and the velocity continuity conditions. To ensure that velocity boundary conditions are met, the first subsidiary constraint mentioned in Appendix B was imposed. Integration and optimization were carried out as discussed in Section 3.1, with the difference that the round-to-oval pass analysis contains only seven unknown coefficients: a_1 to c_2 and V_e.

4. Results

The analysis above was applied to square-to-oval and oval-to-round rolling operations for which other results were available from experiments or analyses discussed in the literature. Comparisons were made between those results and the results acquired from the present study.

4.1. *Square-to-oval pass*

The square-to-oval pass was evaluated against the experiment/analysis carried out by Kennedy (1986). Table 1 outlines the analytical conditions for the square-to-oval pass.

Table 1 – Square-to-oval roll pass data (Kennedy, 1986)

Pass shape	Square-to-oval
Entry cross-section dimension (m)	2.69×10^{-2}
Exit cross-section dimensions (major dim. × minor dim.) (m^2)	$(3.29 \times 2.38) \times 10^{-4}$
Friction factor	0.75
Reduction in area (%)	15
Center distance (m)	0.158×10^{-2}
Roll tangential velocity (m/s)	0.254

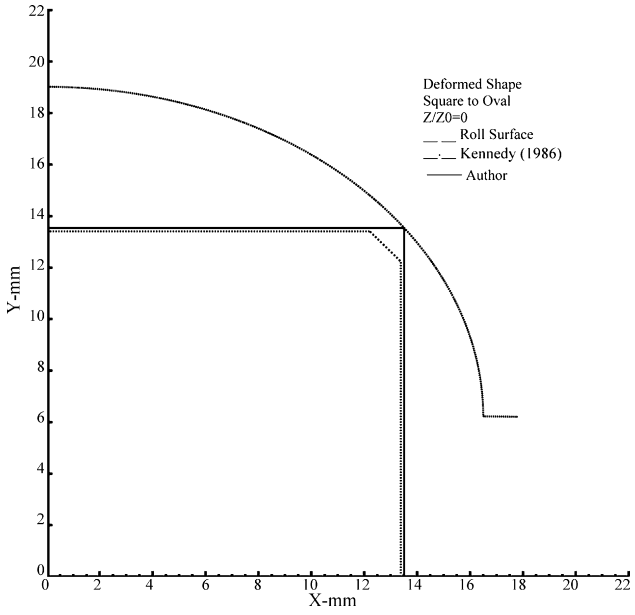


Fig. 6 – Predicted cross-section at $z/z_0 = 0$ (entry) for square-to-oval pass (cross-section found by Kennedy (1986) is superimposed in dashed line).

The material is low carbon steel – AISI 1018 – and rolling was performed at 1090° at which, according to Kennedy (1986), flow stress may be described by:

$$\bar{\sigma} = 77.5\dot{\bar{\varepsilon}}^{0.192}\ \text{MPa} \tag{18}$$

where $\dot{\bar{\varepsilon}}$ is the effective strain rate defined as:

$$\dot{\bar{\varepsilon}} = \left(\frac{2}{3}\dot{\varepsilon}_{ij}.\dot{\varepsilon}_{ij}\right)^{1/2} \tag{19}$$

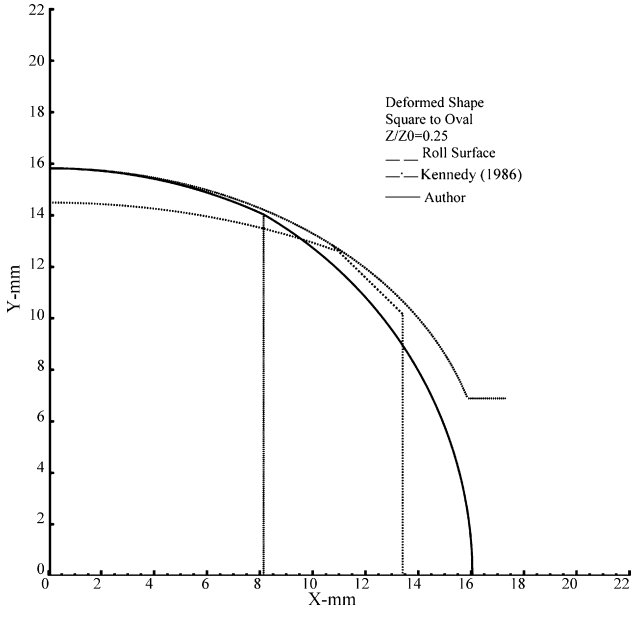


Fig. 7 – Predicted cross-section at $z/z_0 = 0.25$ for square-to-oval pass.

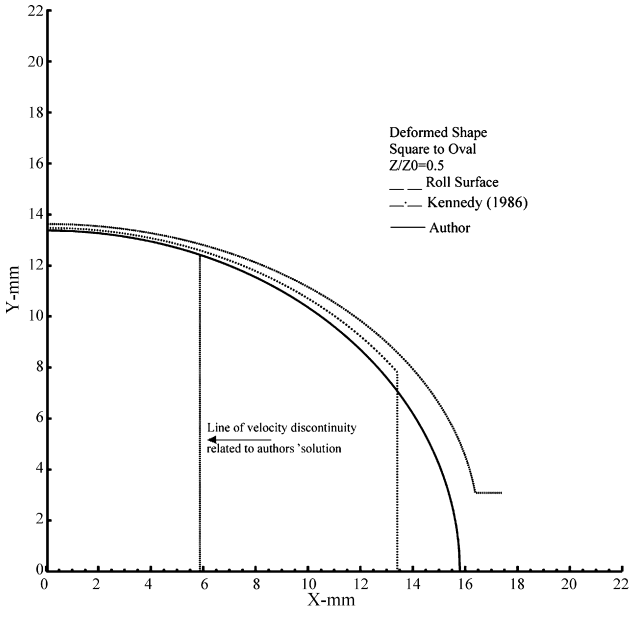


Fig. 8 – Predicted cross-section at $z/z_0 = 0.5$ for square-to-oval pass.

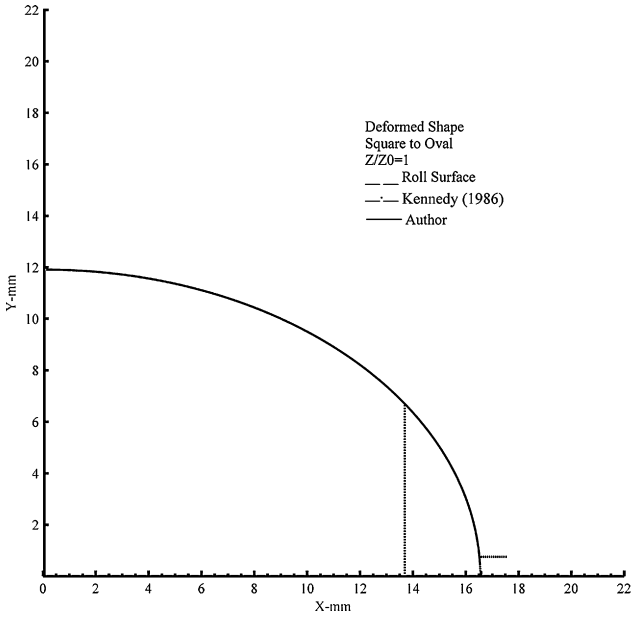


Fig. 10 – Predicted cross-section at $z/z_0 = 1$ (exit) for square-to-oval pass.

Figs. 6–10 illustrate various cross-sections within the deformation zone, from $z/z_0 = 0$ to $z/z_0 = 1$, as predicted by the present method. In each figure, the cross-section shape predicted by Kennedy (1986) has been superimposed.

It can be observed from Figs. 6–10 that in all cases the present study predicts a cross-section closer to the actual roll surface than the work of Kennedy (1986). Due to the streamline formulation, the present analysis predicts a curved, continuous side while the study by Kennedy (1986) predicts a flat side. For the three middle sections, Figs. 7–9, the present study predicts a slight discontinuity, the location of which has

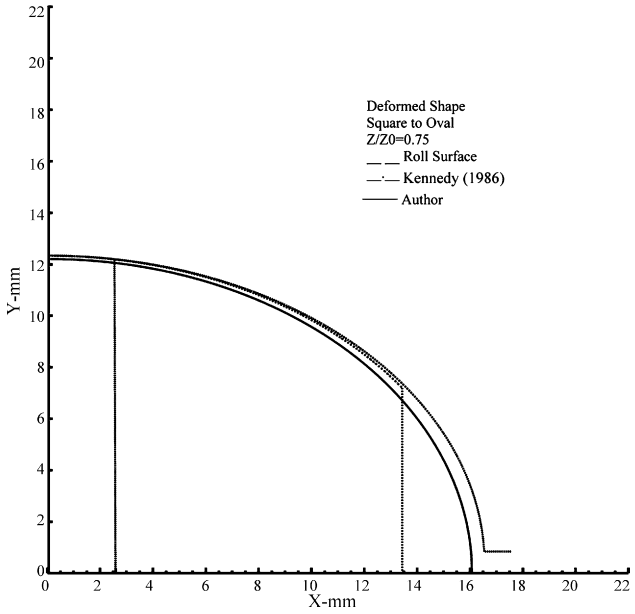


Fig. 9 – Predicted cross-section at $z/z_0 = 0.75$ for square-to-oval pass.

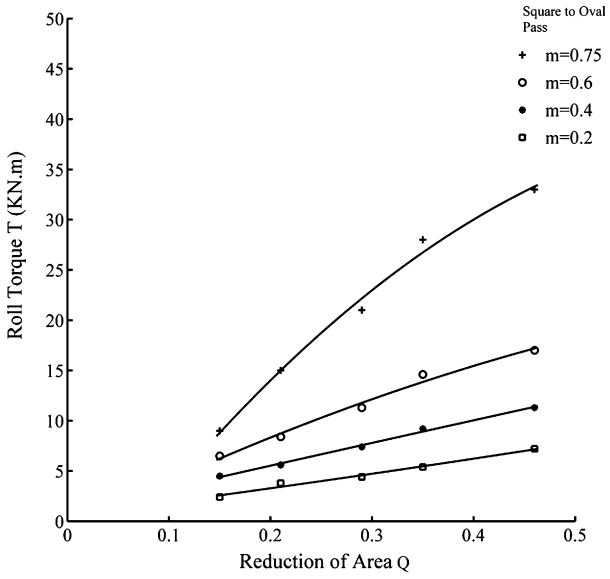


Fig. 11 – Square-to-oval roll torque calculated for various reduction and friction factor values.

Table 2 – Pass data for round-to-oval passes experimented by Wusatowski (1969)

Pass number	1	3
Exit cross-section	Oval	Oval
Exit cross-section dimensions (mm^2)	43×14.8	30.1×10.4
Entry cross-section dimension, Φ (mm)	29.5	20.1
Center distance (mm)	280	280
Reduction of area (%)	26.87	22
Roll rotational speed (rpm)	78	154
Roll gap (mm)	3	2

Table 3 – Average roll pressure and exit velocity results for round-to-oval shape rolling from Wusatowski (1969) experiment, flow-line field solution by Bayoumi (1998) and upper bound method (present analysis)

	Pass 1 (round-to-oval)		
	Friction shear factor (m)	Average roll pressure, P_{av} (MPa)	Exit velocity, V_{ex} (m/s)
Experiment by Wusatowski (1969)	0.374	101.8	1.089
Upper bound (present analysis)	0.374	104.5	1.228
Flow-line field solution by Bayoumi (1998)	0.374	102	1.119

Table 4 – Average roll pressure and exit velocity results for round-to-oval shape rolling from Wusatowski (1969) experiment, flow-line field solution by Bayoumi (1998) and upper bound method (present analysis)

	Pass 3 (round-to-oval)		
	Friction shear factor (m)	Average roll pressure, P_{av} (MPa)	Exit velocity, V_{ex} (m/s)
Experiment by Wusatowski (1969)	0.313	122	2.216
Upper bound (present analysis)	0.313	126.7	2.236
Flow-line field solution by Bayoumi (1998)	0.313	122.2	2.228

been denoted by a straight line. This discontinuity is due to the inevitable approximation when applying the second constraint mentioned in Appendix B.

Roll torque has been calculated by varying roll pass dimensions to incorporate several area reduction values ($Q = 1 - A_2/A_1$) and by applying different frictional conditions; $m = 0.2$, 0.4, 0.6 and 0.75. The effect of friction on roll torque was investigated in Fig. 11.

The same area reduction and friction factor values have been used to calculate the ratio of the average roll pressure to flow stress as shown in Fig. 12. Average roll pressure is obtained by dividing the roll separating force by average contact area. To find the roll separating force, the slab method of analysis suggested by Kennedy (1986) has been adopted. It can be seen from Figs. 11 and 12 that both the roll torque and the ratio of the average roll pressure to flow stress increase with an increase in reduction and friction factor, in effect requiring greater rolling power. Friction factor directly affects the power required to overcome friction forces and thus the total rolling power. The effect of reduction is less explicit, influencing both entry and exit cross-section dimensions, velocity fields and all of the rolling power components.

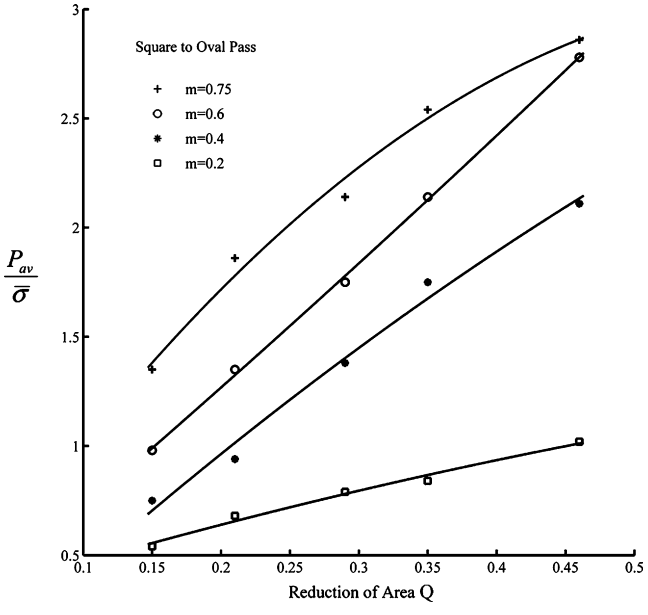


Fig. 12 – The ratio of average roll pressure to flow stress calculated for various reduction and friction factor values.

4.2. *Round-to-oval pass*

The round-to-oval analysis was evaluated against the experiments conducted by Wusatowski (1969) and the flow-line field solution attempted by Bayoumi (1998). The experiments carried out by Wusatowski (1969) include a four pass round–oval–round sequence in which the material, temperature and flow stress data are the same as in Section 4.1. Pass data for the round-to-oval passes analyzed in this study are outlined in Table 2.

Tables 3 and 4 show results for average roll pressure and exit velocity from the present analysis, the flow-line field solution by Bayoumi (1998) and the experiment by Wusatowski (1969).

In both cases, the present upper bound method predicts the average roll pressure and exit velocity with acceptable accuracy, although the amounts predicted here are higher than the flow-line field solution predictions.

5. Summary and conclusions

A method has been proposed to analyze external shape, roll torque, average roll pressure and exit velocity for the rolling of simple geometrical shapes. The method employs the upper bound solution in which numerical methods have been used to optimize the total power required to deform the material and in the process unknown variables have been determined. The upper bound method was based on a new application of parametric curve formulation to define streamlines within the deformation zone. The formulation used simplifying assumptions to enable analysis of the fairly complex metal flow in the rolling operations discussed. Based upon the parametric formulation, a kinematically admissible velocity field was derived and based on this velocity field, an upper bound was established for the power.

Based on the comparison made between the results obtained from this analysis and experimental/analytical studies by other authors, the following conclusions have been made:

- That the shape rolling analysis presented is capable of accurately predicting metal flow patterns, although the sides are curved.
- That the shape rolling analysis presented is capable of accurately predicting roll torque and average roll pressure although, due to the nature of the upper bound method, the analysis predicts slightly higher values.
- That for sections in which the deforming material initiates contact with the roll groove surface toward the outer edge of the entry cross-section, such as the square-to-oval pass, additional parameters are required to allow an accurate representation of the deformation zone in comparison with sections in which initial contact happens toward the inner edges of the entry cross-section, such as the round-to-oval pass. In the former case, the additional surfaces and curves complicate the analysis and slow conversion to the optimum. As a result, the accuracy of the upper bound method is lower than sections which initiate contact toward the inner edges.
- The advantage of the present analysis is that it is very easy to apply to different shapes. In other words since the formulation has a very general form, the user has just to define the final shape of the rolled product and the analysis will be done without any more effort on the part of the user. The formulation presented could be the theoretical base for commercial software for the shape rolling process.
- Process simulation of the shape rolling could easily be done using the formulation presented here. The difficulties arising from the complex nature of three-dimensional flow of material in the design of rolls and pass sequences for shape rolling could be overcome by carrying out the design process in the virtual environment and then using the simulation to analyze the design and identify possible shortcomings or required modifications.
- Unlike numerical methods such as FEM which require a lot of computer time and memory the present method solves the shaped rolling problems in a matter of minutes.
  Also with FEM analysis every time a change occurs in the geometry of a new problem all the pre-processing, processing and post processing must be done all over again while the method presented here is capable of adapting to new geometries without any change in the formulation of the analysis due to its generality.

Appendix A

In tensor notation, the relationship between strain rates and velocities may be written as:

$$\dot{\varepsilon}_{ij} = \frac{1}{2}\left(\frac{\partial V_i}{\partial x_j} + \frac{\partial V_j}{\partial x_i}\right) \tag{A.1}$$

Therefore the incompressibility condition (8) becomes:

$$\dot{\varepsilon}_{ii} = 0 \tag{A.2}$$

Introducing partial differentiation in three dimensions:

$$\frac{\partial V_i}{\partial x_k} = \sum_{j=1}\left(\frac{\partial V_i}{\partial u_j}\cdot\frac{\partial u_j}{\partial x_k}\right) \tag{A.3}$$

or in matrix format:

$$\frac{\partial V_i}{\partial x_k} = \begin{bmatrix} \frac{\partial V_x}{\partial x} & \frac{\partial V_x}{\partial y} & \frac{\partial V_x}{\partial z} \\ \frac{\partial V_y}{\partial x} & \frac{\partial V_y}{\partial y} & \frac{\partial V_y}{\partial z} \\ \frac{\partial V_z}{\partial x} & \frac{\partial V_z}{\partial y} & \frac{\partial V_z}{\partial z} \end{bmatrix} \tag{A.4}$$

$$\frac{\partial V_i}{\partial u_j} = \begin{bmatrix} \frac{\partial V_x}{\partial u} & \frac{\partial V_x}{\partial q} & \frac{\partial V_x}{\partial t} \\ \frac{\partial V_y}{\partial u} & \frac{\partial V_y}{\partial q} & \frac{\partial V_y}{\partial t} \\ \frac{\partial V_z}{\partial u} & \frac{\partial V_z}{\partial q} & \frac{\partial V_z}{\partial t} \end{bmatrix} \tag{A.5}$$

$$\frac{\partial x_k}{\partial u_j} = \begin{bmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial q} & \frac{\partial x}{\partial t} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial q} & \frac{\partial y}{\partial t} \\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial q} & \frac{\partial z}{\partial t} \end{bmatrix} = \begin{bmatrix} f_u & f_q & f_t \\ g_u & g_q & g_t \\ h_u & h_q & h_t \end{bmatrix} = J \tag{A.6}$$

where J is the Jacobian for conversion of coordinates from x, y, z to u, q, t. Therefore:

$$\frac{\partial u_j}{\partial x_k} = J^{-1} = \frac{1}{|J|}\begin{bmatrix} I_{11} & I_{12} & I_{13} \\ I_{21} & I_{22} & I_{23} \\ I_{31} & I_{32} & I_{33} \end{bmatrix} \tag{A.7}$$

where:

$$|J| = \frac{\partial(x, y, z)}{\partial(u, q, t)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial q} & \frac{\partial x}{\partial t} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial q} & \frac{\partial y}{\partial t} \\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial q} & \frac{\partial z}{\partial t} \end{vmatrix} \tag{A.8}$$

Replacing (A.4), (A.5) and (A.7) in (A.3):

$$\begin{bmatrix} \frac{\partial V_x}{\partial x} & \frac{\partial V_x}{\partial y} & \frac{\partial V_x}{\partial z} \\ \frac{\partial V_y}{\partial x} & \frac{\partial V_y}{\partial y} & \frac{\partial V_y}{\partial z} \\ \frac{\partial V_z}{\partial x} & \frac{\partial V_z}{\partial y} & \frac{\partial V_z}{\partial z} \end{bmatrix} = \frac{1}{|J|}\begin{bmatrix} \frac{\partial V_x}{\partial u} & \frac{\partial V_x}{\partial q} & \frac{\partial V_x}{\partial t} \\ \frac{\partial V_y}{\partial u} & \frac{\partial V_y}{\partial q} & \frac{\partial V_y}{\partial t} \\ \frac{\partial V_z}{\partial u} & \frac{\partial V_z}{\partial q} & \frac{\partial V_z}{\partial t} \end{bmatrix}\begin{bmatrix} I_{11} & I_{12} & I_{13} \\ I_{21} & I_{22} & I_{23} \\ I_{31} & I_{32} & I_{33} \end{bmatrix} \tag{A.9}$$

The elements of the J^{-1} matrix can be calculated as follows:

$$\begin{aligned}
I_{11} &= g_q h_t - g_t h_q \\
I_{12} &= -f_q h_t + f_t h_q \\
I_{13} &= f_q g_t - f_t g_q \\
I_{21} &= -g_u h_t + g_t h_u \\
I_{22} &= f_u h_t - f_t h_u \\
I_{23} &= -f_u g_t + f_t g_u \\
I_{31} &= g_u h_q - g_q h_u \\
I_{32} &= -f_u h_q + f_q h_u \\
I_{33} &= f_u g_q - f_q g_u
\end{aligned} \tag{A.10}$$

Differentiating (9) with respect to u, q, t gives:

$$\begin{aligned}
\frac{\partial V_x}{\partial u} &= \frac{(f_{tu}\cdot M + f_t \cdot M_u)h_t - f_t \cdot M \cdot h_{tu}}{h_t^2} \\
\frac{\partial V_x}{\partial q} &= \frac{(f_{tq}\cdot M + f_t \cdot M_q)h_t - f_t \cdot M \cdot h_{tq}}{h_t^2} \\
\frac{\partial V_x}{\partial t} &= \frac{(f_{tt}\cdot M + f_t \cdot M_t)h_t - f_t \cdot M \cdot h_{tt}}{h_t^2} \\
\frac{\partial V_y}{\partial u} &= \frac{(g_{tu}\cdot M + g_t \cdot M_u)h_t - g_t \cdot M \cdot h_{tu}}{h_t^2} \\
\frac{\partial V_y}{\partial q} &= \frac{(g_{tq}\cdot M + g_t \cdot M_q)h_t - g_t \cdot M \cdot h_{tq}}{h_t^2} \\
\frac{\partial V_y}{\partial t} &= \frac{(g_{tt}\cdot M + g_t \cdot M_t)h_t - g_t \cdot M \cdot h_{tt}}{h_t^2} \\
\frac{\partial V_z}{\partial u} &= M_u, \quad \frac{\partial V_z}{\partial q} = M_q, \quad \frac{\partial V_z}{\partial t} = M_t
\end{aligned} \tag{A.11}$$

Expanding (A.9):

$$\begin{aligned}
\frac{\partial V_x}{\partial x} &= \frac{1}{|J|}\left(I_{11}\frac{\partial V_x}{\partial u} + I_{21}\frac{\partial V_x}{\partial q} + I_{31}\frac{\partial V_x}{\partial t}\right) \\
\frac{\partial V_x}{\partial y} &= \frac{1}{|J|}\left(I_{12}\frac{\partial V_x}{\partial u} + I_{22}\frac{\partial V_x}{\partial q} + I_{32}\frac{\partial V_x}{\partial t}\right) \\
\frac{\partial V_x}{\partial z} &= \frac{1}{|J|}\left(I_{13}\frac{\partial V_x}{\partial u} + I_{23}\frac{\partial V_x}{\partial q} + I_{33}\frac{\partial V_x}{\partial t}\right)
\end{aligned}$$

$$\begin{aligned}
\frac{\partial V_y}{\partial x} &= \frac{1}{|J|}\left(I_{11}\frac{\partial V_y}{\partial u} + I_{21}\frac{\partial V_y}{\partial q} + I_{31}\frac{\partial V_y}{\partial t}\right) \\
\frac{\partial V_y}{\partial y} &= \frac{1}{|J|}\left(I_{12}\frac{\partial V_y}{\partial u} + I_{22}\frac{\partial V_y}{\partial q} + I_{32}\frac{\partial V_y}{\partial t}\right) \\
\frac{\partial V_y}{\partial z} &= \frac{1}{|J|}\left(I_{13}\frac{\partial V_y}{\partial u} + I_{23}\frac{\partial V_y}{\partial q} + I_{33}\frac{\partial V_y}{\partial t}\right) \\
\frac{\partial V_z}{\partial x} &= \frac{1}{|J|}\left(I_{11}\frac{\partial V_z}{\partial u} + I_{21}\frac{\partial V_z}{\partial q} + I_{31}\frac{\partial V_z}{\partial t}\right) \\
\frac{\partial V_z}{\partial y} &= \frac{1}{|J|}\left(I_{12}\frac{\partial V_z}{\partial u} + I_{22}\frac{\partial V_z}{\partial q} + I_{32}\frac{\partial V_z}{\partial t}\right) \\
\frac{\partial V_z}{\partial z} &= \frac{1}{|J|}\left(I_{13}\frac{\partial V_z}{\partial u} + I_{23}\frac{\partial V_z}{\partial q} + I_{33}\frac{\partial V_z}{\partial t}\right)
\end{aligned} \tag{A.12}$$

Replacing from (A.10), (A.11) and (A.12) into (8):

$$\begin{aligned}
&\frac{1}{|J|}[(g_q h_t - g_t h_q)\frac{(f_{tu}\cdot M + f_t \cdot M_u)h_t - f_t \cdot M \cdot h_{tu}}{h_t^2} \\
&+(-g_u h_t + g_t h_u)\frac{(f_{tq}\cdot M + f_t \cdot M_q)h_t - f_t \cdot M \cdot h_{tq}}{h_t^2} \\
&+(g_u h_q - g_q h_u)\frac{(f_{tt}\cdot M + f_t \cdot M_t)h_t - f_t \cdot M \cdot h_{tt}}{h_t^2} \\
&+(-f_q h_t + f_t h_q)\frac{(g_{tu}\cdot M + g_t \cdot M_u)h_t - g_t \cdot M \cdot h_{tu}}{h_t^2} \\
&+(f_u h_t - f_t h_u)\frac{(g_{tq}\cdot M + g_t \cdot M_q)h_t - g_t \cdot M \cdot h_{tq}}{h_t^2} \\
&+(-f_u h_q + f_q h_u)\frac{(g_{tt}\cdot M + g_t \cdot M_t)h_t - g_t \cdot M \cdot h_{tt}}{h_t^2} \\
&+(f_q g_t - f_t g_q)M_u + (-f_u g_t + f_t g_u)M_q \\
&+(f_u g_q - f_q g_u)M_t] = 0
\end{aligned} \tag{A.13}$$

or more simply:

$$A(u,q,t)\cdot M + B(u,q,t)\cdot \frac{\partial M}{\partial t} = 0 \tag{A.14}$$

where:

$$\begin{aligned}
A = {} & h_t(g_q f_{tu} h_t - g_q h_{tu} f_t - g_t h_q f_{tu} - g_u h_t f_{tq} + g_u h_{tq} f_t + g_t h_u f_{tq} \\
& + g_u h_q f_{tt} - g_q h_u f_{tt} - f_q g_{tu} h_t + f_q h_{tu} g_t + f_t h_q g_{tu} + f_u g_{tq} h_t \\
& - f_u h_{tq} g_t - f_t h_u g_{tq} - f_u h_q g_{tt} + f_q h_u g_{tt}) + g_t h_q h_{tu} f_t \\
& - g_t h_u h_{tq} f_t - g_u h_q f_t h_{tt} + g_q h_u h_{tt} f_t - f_t h_q h_{tu} g_t \\
& + f_t h_u h_{tq} g_t + f_u h_q h_{tt} g_t - f_q h_u h_{tt} g_t
\end{aligned} \tag{A.15}$$

$$B = h_t[h_t(f_u g_q - f_q g_u) + h_q(f_t g_u - f_u g_t) + h_u(f_q g_t - f_t g_q)] \tag{A.16}$$

Rearranging and integrating (A.14) yields:

$$\int \frac{\partial M}{M} = -\int \frac{A}{B}\partial t \tag{A.17}$$

It is evident from (A.15) and (A.16) that:

$$A = \frac{\partial B}{\partial t} - A_1 \tag{A.18}$$

where

$$A_1 = 2h_{tt}[h_t(f_u g_q - f_q g_u) + h_q(f_t g_u - f_u g_t) + h_u(f_q g_t - f_t g_q)] \tag{A.19}$$

therefore:

$$\frac{A}{B} = \frac{\partial B/\partial t - A_1}{B} \tag{A.20}$$

or

$$\frac{A}{B} = \frac{\partial B/\partial t}{B} - \frac{2h_{tt}}{h_t} \tag{A.21}$$

Replacing (A.21) in (A.17):

$$\int \frac{\partial M}{M} = -\int \left(\frac{\partial B/\partial t}{B} - \frac{2h_{tt}}{h_t}\right)\cdot \partial t \tag{A.22}$$

It follows that

$$\mathrm{Ln}(M) = -\mathrm{Ln}(B) + 2\times \mathrm{Ln}(h_t) + \mathrm{Ln}(C) \tag{A.23}$$

in which C is the constant of integration and a function of the other two parameters u and q;

$$C = C(u,q) \tag{A.24}$$

Rewriting (A.23):

$$M = \frac{h_t^2\cdot C}{B} \tag{A.25}$$

From (A.16):

$$M = \frac{C\cdot h_t^2}{h_t[h_t(f_u g_q - f_q g_u) + h_q(f_t g_u - f_u g_t) + h_u(f_q g_t - f_t g_q)]} \tag{A.26}$$

or

$$M = \frac{C\cdot h_t}{h_t(f_u g_q - f_q g_u) + h_q(f_t g_u - f_u g_t) + h_u(f_q g_t - f_t g_q)} \tag{A.27}$$

Appendix B

The parametric Bezier formulation for the first streamline, e.g. QQ′, may be expressed as follows:

$$\begin{aligned}
\vec{V}_0 &= ustg\varphi\cdot\vec{i} + us\cdot\vec{j}\\
\vec{V}_1 &= a_1 ustg\varphi\cdot\vec{i} + b_1 us\cdot\vec{j} + c_1 z_0\cdot\vec{k}\\
\vec{V}_2 &= a_2 ustg\varphi\cdot\vec{i} + b_2 u\\
&\quad\times\left(y_c - \sqrt{\left(y_c - \frac{b}{a}\sqrt{a^2 - s^2tg^2\varphi}\right)^2 - z_0^2 tg^2\varphi}\right)\cdot\vec{j}\\
&\quad + c_2(z_0 - z_0 utg\varphi)\cdot\vec{k}\\
\vec{V}_3 &= ustg\varphi\cdot\vec{i} + u\\
&\quad\times\left(y_c - \sqrt{\left(y_c - \frac{b}{a}\sqrt{a^2 - s^2tg^2\varphi}\right)^2 - z_0^2 tg^2\varphi}\right)\cdot\vec{j}\\
&\quad + (z_0 - z_0 utg\varphi)\cdot\vec{k}
\end{aligned} \tag{B.1}$$

where $\phi = \pi q/4$ describes the angular position of the first control point $\vec{V}_0$ and the curve may be expressed as:

$$\vec{r}(u,q,t) = (1-t)^3\vec{V}_0 + 3t(1-t^2)\vec{V}_1 + 3t^2(1-t)\vec{V}_2 + t^3\vec{V}_3 \tag{B.2}$$

or in Cartesian form:

$$\vec{r}(u,q,t) = f(u,q,t)\vec{i} + g(u,q,t)\vec{j} + h(u,q,t)\vec{k} \tag{B.3}$$

Velocities derived based on this curve are:

$$\begin{aligned}
M = V_z &= \frac{\pi V_e s^2 u}{4\cos^2(\pi q/4)[(f_q g_u - f_u g_q) - (h_q/h_t)(f_t g_u - f_u g_t) - (h_u/h_t)(f_q g_t - f_t g_q)]}\\
V_x &= \frac{f_t}{h_t}V_z\\
V_y &= \frac{g_t}{h_t}V_z
\end{aligned} \tag{B.4}$$

where V_e is the velocity of billet upon entry to the deformation zone.

The Bezier formulation for the second streamline, e.g. Q′Q″, may be expressed as follows:

$$\begin{aligned}
\vec{V}'_0 &= ustg\varphi\cdot\vec{i}\\
&\quad + u\left(y_c - \sqrt{\left(y_c - \frac{b}{a}\sqrt{a^2 - s^2tg^2\varphi}\right)^2 - z_0^2 tg^2\varphi}\right)\cdot\vec{j}\\
&\quad + (z_0 - z_0 utg\varphi)\cdot\vec{k}\\
\vec{V}'_1 &= [1 + k_1(1-a_2)]ustg\varphi\cdot\vec{i} + [(1 + k_1(1-b_2)]u\\
&\quad\times\left(y_c - \sqrt{\left(y_c - \frac{b}{a}\sqrt{a^2 - s^2tg^2\varphi}\right)^2 - z_0^2 tg^2\varphi}\right)\cdot\vec{j}\\
&\quad + [(1 + k_1(1-c_2)](z_0 - z_0 utg\varphi)\cdot\vec{k}\\
\vec{V}'_2 &= [2k_1(1-a_2) + k_2(1-2a_2) + k_2 a_1]ustg\varphi\cdot\vec{i}\\
&\quad + [[2k_1(1-b_2) + k_2(1-2b_2) + 1]u\,(y_c\\
&\quad - \sqrt{\left(y_c - \frac{b}{a}\sqrt{a^2 - s^2tg^2\varphi}\right)^2 - z_0^2 tg^2\varphi}\Bigg) + k_2 b_1 us\Bigg]\cdot\vec{j}\\
&\quad + [[2k_1(1-c_2) + k_2(1-2c_2) + 1](z_0 - z_0 utg\varphi)\\
&\quad + k_2 c_1 z_0]\cdot\vec{k}\\
\vec{V}'_3 &= u(am)\cdot\vec{i} + u(b\sqrt{1-m^2})\cdot\vec{j} + z_0\cdot\vec{k}
\end{aligned} \tag{B.5}$$

where $m = \sin(\pi/4tg\varphi)$ and k_1 and k_2 are additional parameters introduced to maintain slope and curvature at the junction of the two streamlines, e.g. QQ′ and Q′Q″. Equations (B.1) and (B.5) ensure that there is no velocity discontinuity at the junction point.

The second curve may be expressed in parametric terms as follows:

$$\vec{r}'(u,q,t) = (1-t)^3\vec{V}'_0 + 3t(1-t^2)\vec{V}'_1 + 3t^2(1-t)\vec{V}'_2 + t^3\vec{V}'_3 \tag{B.6}$$

or in Cartesian coordinates:

$$\vec{r}'(u,q,t) = f'(u,q,t)\vec{i} + g'(u,q,t)\vec{j} + h'(u,q,t)\vec{k} \tag{B.7}$$

The velocity field in this zone has been found to be:

$$M' = V'_z = \frac{\pi V_e s^2 u}{4\cos^2(\pi q/4)[(f'_q g'_u - f'_u g'_q) - (h'_q/h'_t)(f'_t g'_u - f'_u g'_t) - (h'_u/h'_t)(f'_q g'_t - f'_t g'_q)]}$$
$$V'_x = \frac{f'_t}{h'_t} V'_z$$
$$V'_y = \frac{g'_t}{h'_t} V'_z \tag{B.8}$$

Streamlines in the second half of the deformation zone, e.g. SS', may be described in parametric Bezier formulation as follows:

$$\vec{V}''_0 = us \cdot \vec{i} + ustg\varphi \cdot \vec{j}$$
$$\vec{V}''_1 = a'_1 us \cdot \vec{i} + b'_1 ustg\varphi \cdot \vec{j} + c'_1 z_0 \cdot \vec{k}$$
$$\vec{V}''_2 = a'_2 u(a\sqrt{1-m^2}) \cdot \vec{i} + b'_2 u(bm) \cdot \vec{j} + c'_2 z_0 \cdot \vec{k} \tag{B.9}$$
$$\vec{V}''_3 = u(a\sqrt{1-m^2}) \cdot \vec{i} + u(bm) \cdot \vec{j} + z_0 \cdot \vec{k}$$

In parametric terms:

$$\vec{r}''(u,q,t) = (1-t)^3\vec{V}''_0 + 3t(1-t^2)\vec{V}''_1 + 3t^2(1-t)\vec{V}''_2 + t^3\vec{V}''_3 \tag{B.10}$$

and in Cartesian coordinates:

$$\vec{r}''(u,q,t) = f''(u,q,t)\vec{i} + g''(u,q,t)\vec{j} + h''(u,q,t)\vec{k} \tag{B.11}$$

The velocity field in the second half of the deformation zone in a square-to-oval pass has been found as:

$$M'' = V''_z = \frac{\pi V_e s^2 u}{4\cos^2(\pi q/4)[(f''_u g''_q - f''_q g''_u) + (h''_q/h''_t)(f''_t g''_u - f''_u g''_t) + (h''_u/h''_t)(f''_q g''_t - f''_t g''_q)]}$$
$$V''_x = \frac{f''_t}{h''_t} V''_z$$
$$V''_y = \frac{g''_t}{h''_t} V''_z \tag{B.12}$$

The relations for velocities developed so far satisfy both the incompressibility and velocity continuity conditions. To ensure that velocity boundary conditions are met, a subsidiary constraint was imposed; streamlines located on the roll–workpiece interface must also be located on the roll surface. In mathematical terms, this translates into the condition that where $u = 1$, the parametric expression of the curve must satisfy the three-dimensional roll surface Eq. (15).

A second constraint was imposed to preclude velocity discontinuities between streamlines located on the boundary between the two halves of the entry cross-section. In mathematical terms, this means that where $q = 1$, $\vec{r}(u,q,t) = \vec{r}''(u,q,t)_{q=1}$.

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